

Ado. Net

Agenda

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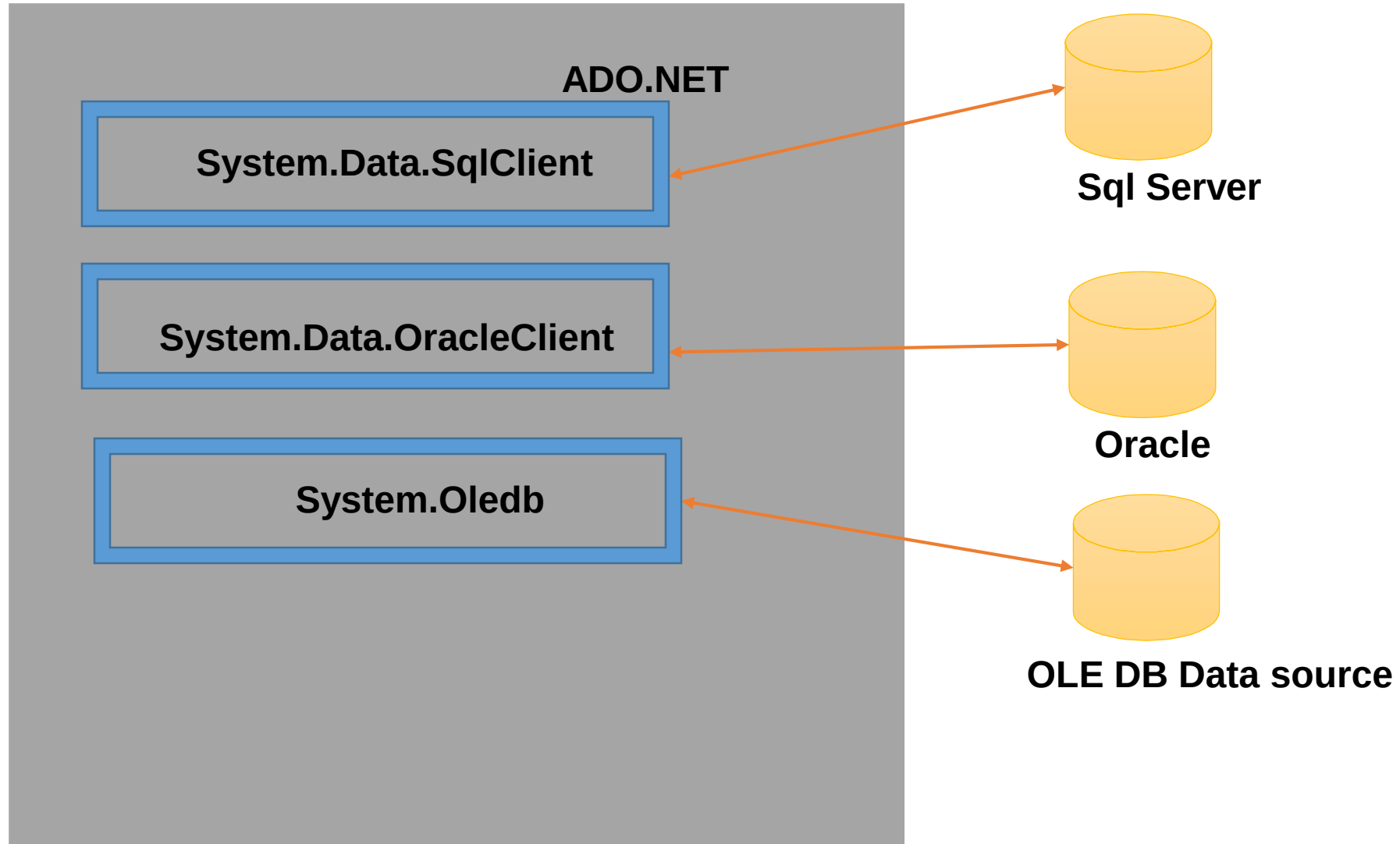
Hands-On Assignments

What is ADO.NET?

- ADO.NET is a set of Framework classes which allows to interact with Data sources
- **Data Sources includes following**
 - Sql Server
 - Oracle
 - MySQL
 - Access
 - Excel
 - Xml Files etc...



.NET Data Providers



Models in ADO.NET

Connected Architecture

To perform CRUD operation, database connection has to exist

Disconnected Architecture

All operations are performed offline, all offline changes are updated as a bulk to database

ADO.NET Architecture

Components of ADO.NET Object model

— .NET Data Providers

- SqlClient
- OracleClient
- Oledb

— Connected Model API's

- SqlConnection
- SqlCommand
- SqlDataReader

— Disconnected Model API's

- SqlDataAdapter
- DataSet

Steps in connected Data Architecture

Imports namespace

Create connection object

Open connection

Create command object

Execute Command

Extract the Results

Close the connection

Connection class

Constructor

- Connection(connection String)
- Connection string
 - “provider = ;server= ; user id=; password=; “

Methods

- Open()
- Close()

Command class

Constructor

- `Command(command String, connection object)`

Methods

- `ExecuteReader()` returns `DataReader`
- `ExecuteNonQuery()` returns `int`
- `ExecuteScalar()` returns `Object`
- `ExecuteXmlReader()` returns `XmlReader`

Invoking Stored Procedures

- Not only simple DML operations, we can also invoke stored procedures using command class
- In order to invoke stored Procedure following steps need to be followed
 1. Set `SqlCommand.CommandText` with "Name of storeProcedure"
 2. Set `SqlCommand.CommandType` to `CommandType.StoredProcedure`
 3. Execute the storedprocedure like any other DML/DQL command using either `ExecuteReader`, `ExecuteScalar` or `ExecuteNonQuery` methods

Disconnected Model

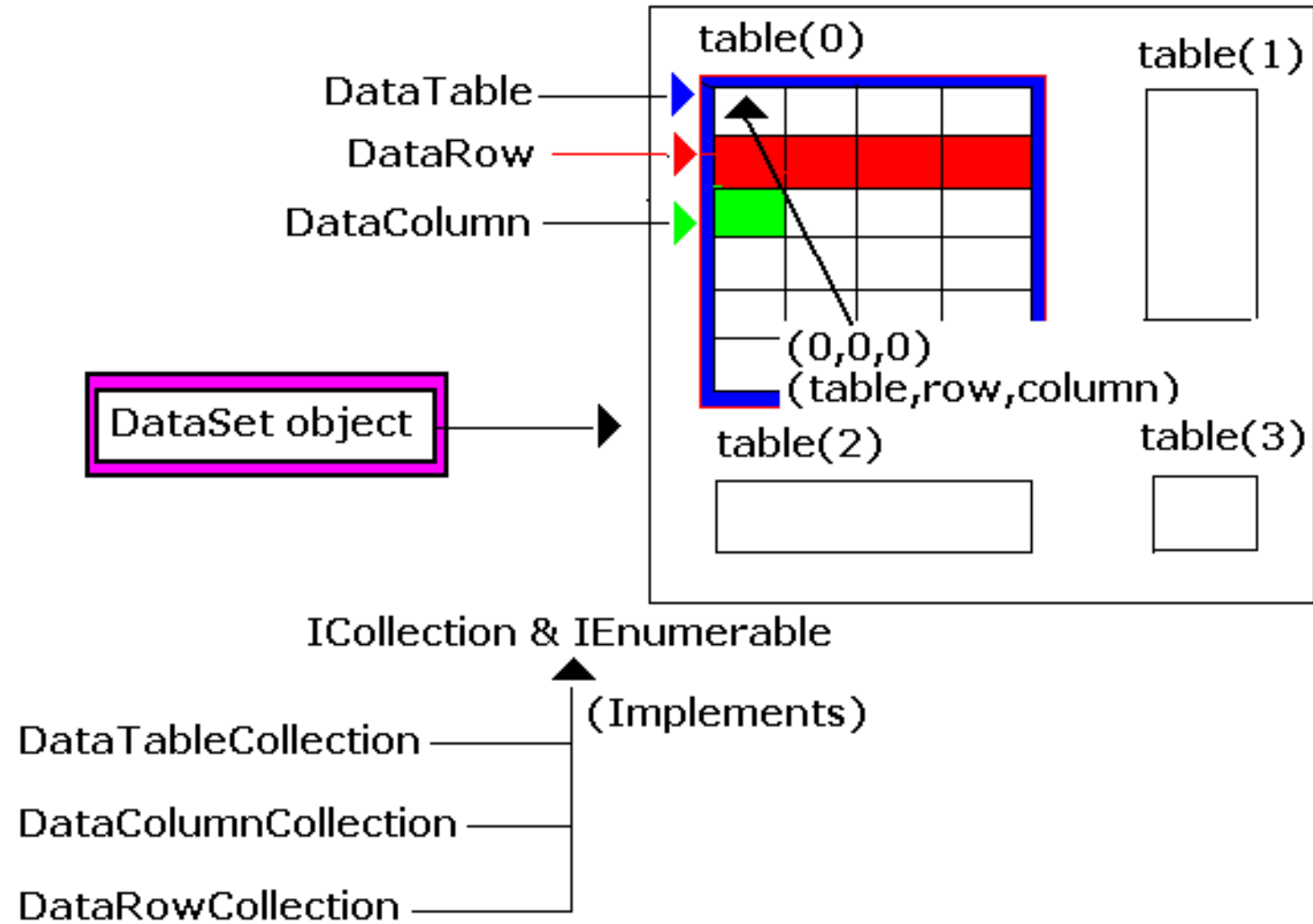
DataSet

DataSet a Collection of

- Tables
- Relations (Constraints)
- Extended properties (Custom Properties)

DataSet can also have views for the tables

DataSet Architecture



Comparison DataReader & DataSet

Advantages and Limitations

DataSet

Advantages

- Its magnitude is closer to the database
- It offers index based access
- It allows update ,delete also
- Always works in **disconnected mode**
- Can read data from xml file too

Limitations

- Slow
- Each piece of data in the DataSet is by default of type object.

DataReader

• Limitations

- It is singly
- It is forward only, provides sequential access
- It is read only
- Always works in **connected mode**
- Not xml enabled

• Advantages

- Fast
- Each piece of data in the DataReader is of specific type .

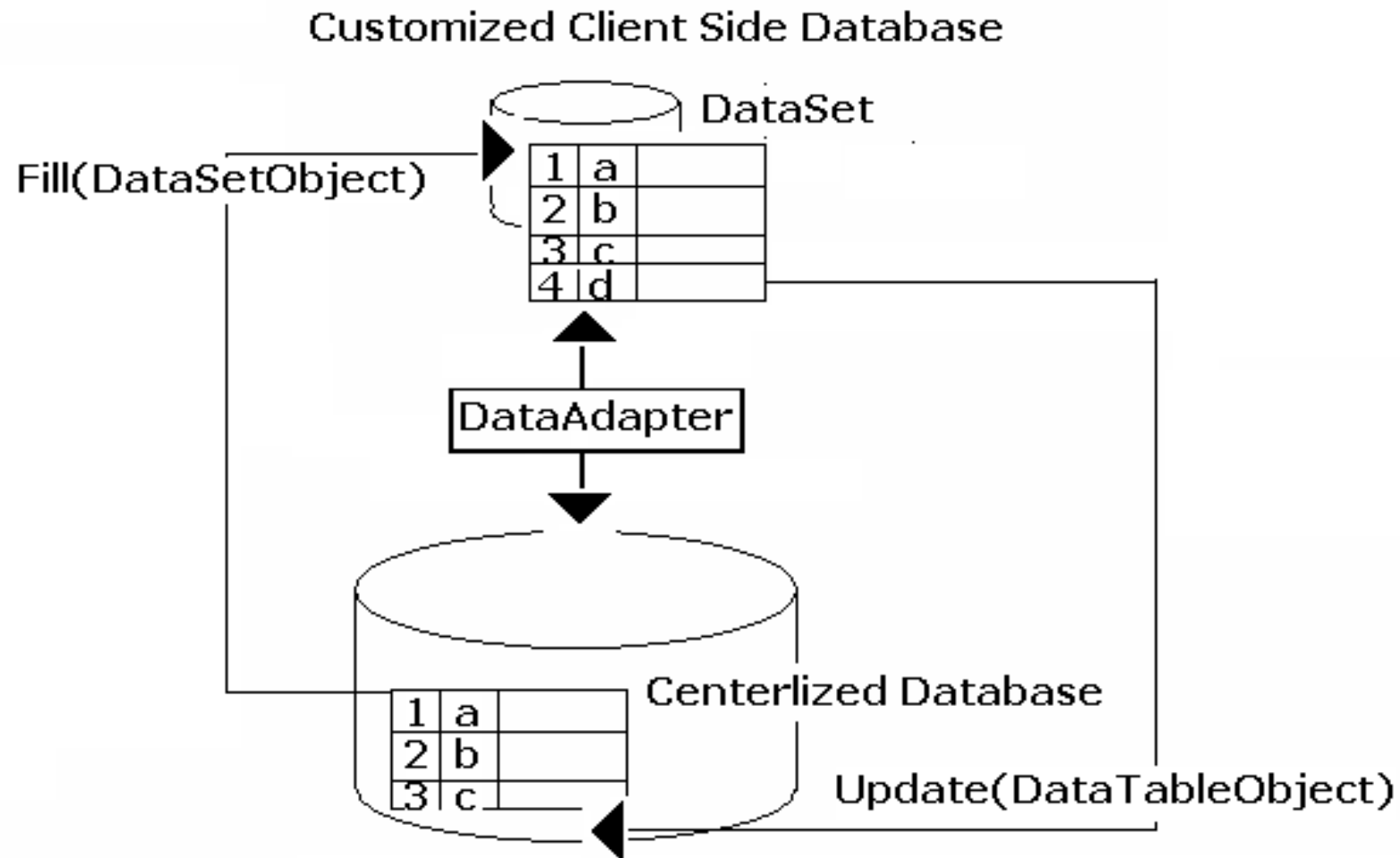
Implementing Disconnected Data Architecture

The Core elements

- DataSet
 - Client side in-memory representation of data
 - Generic to any type of DataProvider
- DataAdapter
 - Back bone of disconnected data architecture in ADO.NET
 - Mediator between centralized database and client side customized database i.e. DataSet

Disconnected Data Architecture

- High level view



XML Features in ADO.NET

In the connected data architecture

- ExecuteXmlReader()
- Supported only by SQL Server

In the disconnected data architecture

- ReadXml()
 - Reading an xml file into DataSet object
- WriteXml()
 - Writing an xml file out of the contents of DataSet

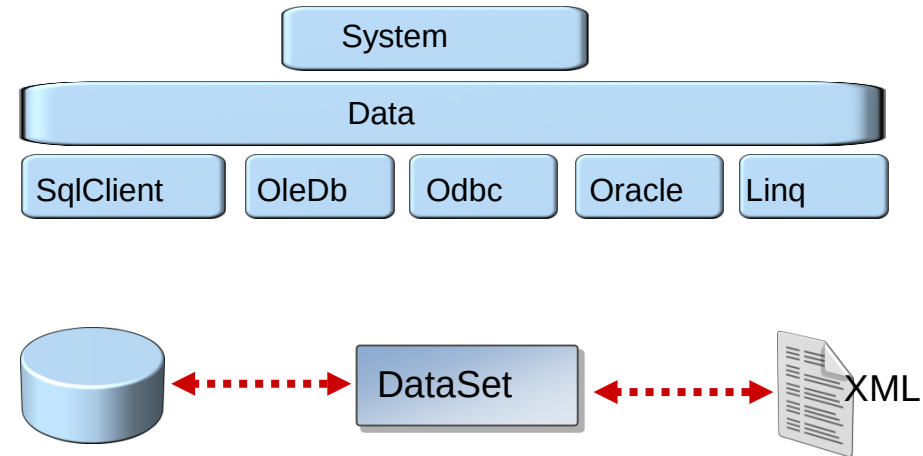
ADO.NET 4.0 Technologies and Techniques

ADO.NET 4.0 Application Programming Interface (API)

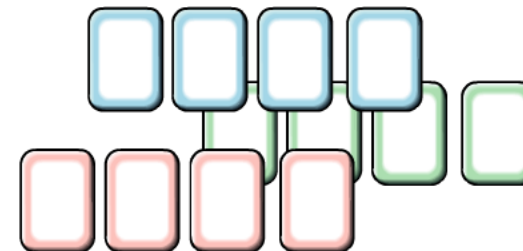
Disconnected data access

Language-Integrated Query (LINQ)

ADO.NET Entity Framework



```
Dim q = From c In db.Customers _  
        Where c.City = "London"  
        Select c
```



Mapping SQL Server Data Types to .NET Framework Types

char, varchar, nchar, nvarchar	String, Char()
varchar(max), nvarchar(max)	String, Char()
int	Integer
decimal, numeric, money, smallmoney	Decimal
float	Double
bit	Boolean
date, datetime, smalldatetime	DateTime
timestamp, binary, varbinary(max)	Byte()